# The Universal/Coulomb Partition of the GeoVac Spectral Triple
# What belongs to angular momentum, what belongs to $-Z/r$, and why the difference matters

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The GeoVac framework [19, 23] is a discrete spectral triple in the sense of Connes [1] and the graph-spectral-triple lineage of Marcolli and van Suijlekom [3]: an algebra $\mathcal{A}$ of functions on the Fock-projected discrete graph, a Hilbert space $\mathcal{H}$ of scalar and spinor states, and a Dirac operator $D$ inherited from the Camporesi–Higuchi spectrum on $S^3$. This paper identifies a clean structural decomposition of the framework's results into two disjoint sectors:

- A *universal sector*, depending only on $\mathcal{A}$ and the angular structure of the basis. The angular sparsity theorem of Paper 22 [27] — the angular ERI density is potential-independent (global-$M_L$ Coulomb $D = 6.06\%$ at $l_{\max} = 3$, the density the composed pipeline realizes; the stricter pair-diagonal restriction gives $D_{\mathrm{pd}} = 1.44\%$), bit-identical across Coulomb, harmonic oscillator, Woods–Saxon, square well, and Yukawa potentials — is the canonical example.
- A *potential-specific sector*, depending on the choice of $D$. The Fock projection rigidity theorem of Paper 23 [28] ($S^3$ projection unique to $-Z/r$ by SO(4) symmetry) and the Bargmann–Segal rigidity theorem of Paper 24 [29] (the 3D isotropic harmonic oscillator is the unique central potential whose spectrum arises from the Euler operator on $H^2(S^5)$ restricted to symmetric SU(3) irreps; its converse half rests on an imported result Paper 24 flags as unverified) jointly establish the partition's sharpness, the second subject to that caveat.

The two rigidity theorems sit on opposite sides of an operator-order divide: Coulomb uses a second-order Riemannian Laplacian, the HO a first-order complex Euler operator. The divide does not decide where $\pi$ enters: both label sets are $\pi$-free, and on both sides $\pi$ enters through a measure and through spectral sums at even argument; in the Kustaanheimo–Stiefel frame the Coulomb problem is itself a constrained oscillator with a linear projection (Paper 24). The $S^5$ gauge-structure result of Paper 24 [29] supplies the standing Coulomb/HO discriminator, Wilson gauge with natural matter coupling, decided by representation theory (on $S^3 = \mathrm{SU}(2)$ matter and links share a representation, on $S^5$ matter coupling needs a Clebsch–Gordan intertwiner). Calibration $\pi$, listed here as a second discriminator until 2026-10-19, is withdrawn. (A third, the spectrum-computing role of $L_0$, [retracted 2026-09-26: coulomb-ho-layer1-and-three-discriminators] was listed here until 2026-09-25; it is withdrawn, because the graph Laplacian on $S^3$ does not converge to the $S^3$ Laplacian and is spectrally inert there as on $S^5$.) We frame the partition in spectral-triple language and identify which features of GeoVac transfer to non-Coulomb systems for free, which require system-specific rederivation, and what this implies for nuclear, oscillator, and gauge-theoretic extensions of the framework.

## I. INTRODUCTION

The GeoVac framework [16, 17, 19] builds quantum-mechanical Hamiltonians on natural geometries over a packed lattice of quantum-number labels. For the hydrogen atom, the label set $(n, l, m)$ is exactly the index set of the $S^3$ harmonics, on which the continuum $S^3$ Laplace–Beltrami operator acts as the diagonal $SO(4)$ Casimir, with spectrum $\lambda_n = -(n^2 - 1)$ (the lattice's edge Laplacian $D - A$ is *not* a discretization of that operator and does not converge to it: it preserves $l$ and splits into $n_{\max}$ disconnected blocks, Paper 0 §VI); this spectrum recovers the hydrogenic Rydberg series under Fock's stereographic projection $p_0 = \sqrt{-2E}$ from $S^3$ to momentum space [7, 8]. The framework has been extended to multi-electron atoms [21], diatomics [22, 24], polyatomics [25], nuclear shell model Hamiltonians [28], and the three-dimensional isotropic harmonic oscillator [29]. In each case the same combinatorial machinery — Wigner $3j$ symbols, Gaunt selection rules, $jj$-coupled angular reduction — produces a sparse one- and two-body Hamiltonian with the same structural features. Paper 22's central observation [27] is that this sparsity is not particular to any of these systems: it is a property of the angular momentum algebra and persists across all spherical fermion bases, regardless of the radial potential.

The same machinery does not produce identical spectra or projections. Above the label level, $\pi$ enters both sides. On $S^3$ the kinetic calibration $\kappa = -1/16$ is itself rational (a matched scale on the graph's extremal eigenvalue; the bridge to the $\pi$-laden continuum is Fock's energy shell $p_0^2 = -2E$, not $\kappa$), its value *coinciding* with the $1/16$ of the Fock projection Jacobian as an Observation, not a derivation (Paper 7); the $\alpha$-combination $K = \pi(B + F - \Delta)$ [18] carries an explicit factor of $\pi$, a chosen prefactor (Paper 2), and the spectral zeta values appear at even integer arguments via the Jacobi-theta inversion formula [26, 31]. The 3D harmonic oscillator

discretized via Bargmann–Segal on $S^5$ is bit-exactly $\pi$-free in exact rational arithmetic at every finite $N_{\max}$ [29]: every adjacency, every dipole transition amplitude, every spectral eigenvalue is a rational number or finite algebraic extension. So is the $S^3$ label set. Above that level the oscillator meets $\pi$ the same two ways: the $\pi^{-3}$ of the Gaussian Bargmann measure, and $\pi^2$ in its spectral zeta values at even argument (Paper 24).

These two observations — universal angular sparsity and potential-specific spectra and projections — point at a structural decomposition of the GeoVac framework that, once stated explicitly, clarifies what the framework does and does not transfer across physical systems. The decomposition is the standard separation of a spectral triple $(\mathcal{A}, \mathcal{H}, D)$ into its algebra $\mathcal{A}$ and its Dirac operator $D$, in the sense of Connes [1] and the graph-spectral-triple framework of Marcolli and van Suijlekom [3]. In this language, angular sparsity belongs to $\mathcal{A}$ (it is a selection rule on which products of basis functions are nonzero, independent of any operator); rigidity, the projection, and the particular spectrum belong to $D$ (they encode the choice of potential and projection).

The contribution of this paper is to state the partition explicitly, give it a name, and trace its consequences. Its scientific content is small: every theorem cited is established in a previous paper of this series. Its organizational content is large: the partition explains *which* GeoVac results are universal across spherical fermion problems and *which* are particular to the Coulomb potential, and it provides a checklist for extending the framework to new potentials without re-deriving universal results.

The paper is structured as follows. Section II introduces the GeoVac framework as a discrete spectral triple that is an *instance* of the Marcolli–van Suijlekom 2014 gauge-network construction (with the Perez-Sanchez 2025 correction) at the data level and a structural extension of that framework via the forced-graph claim (Bertrand $\times$ Fock projection) and the master Mellin engine. Section III restates Paper 22's angular sparsity theorem as the canonical content of the universal sector. Section IV reviews the two known potential-specific rigidity theorems (Fock and Bargmann–Segal) and shows that they sit on opposite sides of an operator-order divide. Section V records where $\pi$ enters on each side and why the operator-order divide does not separate them. Section VI reports the Sprint 5 result that the Coulomb/HO asymmetry has one standing discriminator of the three reported here (within the catalogue of Paper 24 [29], seven layers catalogued, five standing; the spectrum-computing role of $L_0$ and calibration $\pi$ are withdrawn, 2026-09-25 and 2026-10-19), and gives the predictive checklist that follows. Section VII states what features of GeoVac transfer to non-Coulomb systems for free and what does not. Section IX closes with open questions about the extension to non-central potentials and gauge sectors.

**Scope of contribution.** This paper is not a new theorem. It is a structural framing of existing results [27–29, 33] in spectral-triple language. No new computations are performed. The framing is intended to clarify which existing results extend without modification to non-Coulomb systems and which require explicit recomputation, and to place the GeoVac framework in the graph-spectral-triple lineage of Connes, Marcolli, and van Suijlekom [1, 3, 4]. The dual-description rule of the framework — in particular that the Fock equivalence between hydrogen and the continuum Laplace–Beltrami operator on $S^3$, and the identification of the graph's nodes with that operator's labels (on which it is the diagonal $SO(4)$ Casimir), are mathematical equivalences and not ontological claims about which description is more fundamental — is preserved throughout. (An earlier version called the identification of the discrete graph *with* the Laplace–Beltrami operator a proven equivalence; the edge Laplacian does not converge to it, corrected 2026-09-25.)

## II. SETUP: GEOVAC AS A DISCRETE SPECTRAL TRIPLE

A spectral triple in the sense of Connes [1] is a data triple

$$(\mathcal{A}, \mathcal{H}, D)$$

where $\mathcal{A}$ is a $*$-algebra represented as bounded operators on a Hilbert space $\mathcal{H}$, and $D$ is a self-adjoint operator on $\mathcal{H}$ with compact resolvent and bounded commutators $[D, a]$ for all $a \in \mathcal{A}$. In the commutative case $\mathcal{A} = C^\infty(M)$ for a closed Riemannian spin manifold $M$ and $D$ is the Dirac operator on $M$, the triple recovers the geometry of $M$ via Connes' reconstruction theorem. In the noncommutative case, the triple is treated as defining the geometry, and the spectral action principle of Chamseddine and Connes [2] extracts physics by computing $\mathrm{Tr}\, f(D^2/\Lambda^2)$ for suitable test functions $f$ and cutoffs $\Lambda$.

Marcolli and van Suijlekom [3] introduced *gauge networks*: a discrete variant of the spectral triple where $\mathcal{A}$ is supported on the vertices of a graph, $\mathcal{H}$ is a finite-dimensional vector space (one fiber per vertex), and the gauge connection lives on the edges. Their main result is that the Chamseddine–Connes spectral action on a gauge network coincides with the Wilson lattice gauge action of the underlying graph; the Perez-Sanchez correction [5] clarifies that the continuum limit recovers Yang–Mills without Higgs.

The GeoVac framework is an instance of this construction at the data level. Specifically:

**$\mathcal{A}$::** the $*$-algebra of functions on the discrete Fock-projected graph, with vertices labeled by quantum numbers $(n, l, m)$ for the scalar sector and $(n, \kappa, m_j)$ for the spinor sector [19]. Multiplication is pointwise; the involution is complex conjugation. $\mathcal{A}$ acts on $\mathcal{H}$ by diagonal multiplication operators $a_v \cdot \psi_v = (a \cdot \psi)_v$.

$\mathcal{H}$**::** the finite-dimensional Hilbert space of scalar (or spinor) wavefunctions on the graph. Decomposes as $\mathcal{H} = \mathcal{H}^{\text{scalar}} \oplus \mathcal{H}^{\text{spinor}}$ with the spinor sector supplied by the Camporesi–Higuchi construction [31].

$D$**::** the Dirac operator on $\mathcal{H}^{\text{spinor}}$ with eigenvalues $|\lambda_n^{\text{D}}| = n + 3/2$ and degeneracies $g_n^{\text{D}} = 2(n+1)(n+2)$, restricted to a finite cutoff $n \le n_{\max}$ [31]. The squared Dirac operator $D^2$ acts on the scalar sector with the Lichnerowicz formula [6]

$$D^2 = \Delta_{\text{LB}} + \tfrac{R_{\text{scalar}}}{4} = \Delta_{\text{LB}} + \tfrac{3}{2}$$

on the unit $S^3$, recovering the eigenvalues $|\lambda_n|^2 = (n+3/2)^2 = n(n+2) + 9/4$.

The gauge structure of Paper 30 [33] enters as inner derivations of an almost-commutative extension $\mathcal{A} \otimes M_n(\mathbb{C})$, with its compact $U(1)$ (maximal-torus) section corresponding to $n = 1$ and its SU(2) Wilson theory to the $n = 2$ full non-abelian slice.

**The decomposition.** Once the data $(\mathcal{A}, \mathcal{H}, D)$ are fixed, every observable in the framework is a function of these three pieces and a finite list of auxiliary structures (the cutoff $n_{\max}$, the gauge group, the test function $f$ for the spectral action). The partition we develop in this paper observes that the GeoVac results split cleanly along the $\mathcal{A}$ versus $D$ line:

- Results that depend only on $\mathcal{A}$ (and the angular decomposition of its basis) are *universal across central potentials*: they hold for any spectral triple sharing the same algebra and basis, regardless of which Dirac operator is chosen.
- Results that depend on the specific form of $D$ (or equivalently, on the radial potential $V(r)$) are *potential-specific*: they hold for the Coulomb problem (or the harmonic oscillator, or a Woods–Saxon nucleus) but not for arbitrary central potentials.

The decomposition is not novel as a piece of spectral triple formalism — it is the standard $\mathcal{A}/D$ split. The content of this paper is to identify which GeoVac theorems live on each side and what this implies for the framework's extension to new systems.

## III. THE UNIVERSAL SECTOR: ANGULAR SPARSITY

The canonical content of the universal sector is the angular sparsity theorem of Paper 22 [27]. We restate it in spectral-triple language to make the partition framework explicit.

### A. Restatement of the angular sparsity theorem

Let $(\mathcal{A}, \mathcal{H}, D)$ be a GeoVac spectral triple in which $\mathcal{H}$ is decomposed in the spherical-harmonic basis $|n, l, m\rangle$ and the rotationally invariant two-body operator $\hat{V}_{12} = V(|\mathbf{r}_1 - \mathbf{r}_2|)$ admits the Legendre expansion

$$V(|\mathbf{r}_1 - \mathbf{r}_2|) = \sum_{k \ge 0} v_k(r_1, r_2)\, P_k(\cos\theta_{12}).$$

Then the two-body matrix elements factorize [27] as

$$\langle n_1 l_1 m_1, n_2 l_2 m_2 \,|\, \hat{V}_{12} \,|\, n_3 l_3 m_3, n_4 l_4 m_4 \rangle = \sum_k c^k(l_1 m_1; l_3 m_3)\, c^k(l_2 m_2; l_4 m_4)\, R^k(n_1 l_1, n_2 l_2; n_3 l_3, n_4 l_4),$$

where the Gaunt coefficients $c^k$ are pure functions of $(l, m)$ quantum numbers and the radial integrals $R^k$ depend on $V$. The zero structure of the matrix-element tensor is determined by the Gaunt selection rules alone:

- parity: $l_1 + l_3 + k$ and $l_2 + l_4 + k$ both even,
- triangle inequalities $|l_1 - l_3| \le k \le l_1 + l_3$ and $|l_2 - l_4| \le k \le l_2 + l_4$,
- magnetic conservation $m_1 + m_2 = m_3 + m_4$.

None of these conditions reference $V$, $R^k$, or any radial quantum number. Theorem 2 of Paper 22 then states that the ERI density is a function only of the angular momentum cutoff. The composed pipeline realizes the physical global-$M_L$ Coulomb density $D$ (selection rule $m_a + m_b = m_c + m_d$; Paper 22 Table):

$$D(l_{\max}) = \begin{cases} 14.84\% & l_{\max} = 1 \\ 8.52\% & l_{\max} = 2 \\ 6.06\% & l_{\max} = 3 \\ 4.83\% & l_{\max} = 4 \\ 3.99\% & l_{\max} = 5 \end{cases} \tag{1}$$

The stricter *pair-diagonal* ($m$-decoupled) restriction $D_{\text{pd}}$ (per-pair $m_a = m_c$, $m_b = m_d$) is $\sim 4\times$ smaller ($7.81/2.76/1.44/0.90/0.62\%$ at $l_{\max} = 1 \ldots 5$); it is a subpattern of the Gaunt zeros, not the density the pipeline realizes. Paper 22 verified the potential-independence numerically across five potentials (Coulomb, harmonic oscillator, Woods–Saxon, square well, Yukawa) at $l_{\max} = 2$, finding bit-identical zero patterns ($D = 8.52\%$ on the

matched 18-orbital set) in every case.

### B. Spectral-triple reading

In the partition framework, the angular sparsity theorem is a statement about the algebra $\mathcal{A}$ and the SO(3) representation structure of its basis decomposition. The set of nonzero matrix elements

$$\mathcal{N}(\mathcal{A}, l_{\max}) = \{(a,b,c,d) \in \mathcal{A}^{\otimes 4} : \langle ab|\hat{V}_{12}|cd\rangle \neq 0\}$$

is determined by the Gaunt triangle and parity conditions, which depend only on the basis labels $(l, m)$ and not on $D$. Equivalently, $\mathcal{N}$ is a function of $\mathcal{A}$ alone, in the precise sense that any change of $D$ (i.e., any change of radial potential) that preserves the spherical-harmonic basis decomposition of $\mathcal{H}$ leaves $\mathcal{N}$ unchanged.

This is the formal sense in which angular sparsity is universal. The spinor extension developed in Paper 22's spinor-block angular sparsity section [27] carries the same property: the $jj$-coupled angular reduction [13] preserves the radial-angular factorization and the spinor ERI density $d_{\text{spinor}}(l_{\max})$ depends only on $l_{\max}$, not on the choice of $D$. Thus the universal sector includes both scalar and spinor sparsity bounds.

### C. What the universal sector contains

The universal sector encompasses every result of the GeoVac framework that can be derived from the algebra $\mathcal{A}$, the angular decomposition of $\mathcal{H}$, and the Wigner $3j$ algebra, without invoking the specific spectrum of $D$. This includes:

- Angular sparsity bounds: Eq. (1) above.
- The composed-block Pauli count of Paper 14 [23], exactly linear in the qubit count across molecules at fixed basis ($N_{\text{Pauli}} = 27.90 \times Q$; the retired pair-diagonal rule gave $O(Q^{2.5})$), which depends on the block-diagonal structure of two-body matrix elements in a composed orbital basis and the Gaunt selection rules. Verified across the 35 composed molecules in the GeoVac library.
- Selection rules for transitions, oscillator strengths, and multipole moments (the Wigner–Eckart content of any operator expanded in spherical tensor components).
- Block-diagonal structure of two-body interaction tensors in the composed architecture [25].
- The $jj$-coupled qubit encoding sparsity for relativistic composed Hamiltonians [23] §V (Tier 2 spinor extension).

Each of these results was originally derived in the Coulomb context, but the derivation never used a Coulomb-specific feature of $D$ beyond the spherical-harmonic basis decomposition. They transfer to any new central potential without re-derivation, contingent only on the new system having the same basis — except the composed Pauli coefficient 27.90, which Paper 22 (Corollary 1) measures on the Coulomb library and only expects to carry over.

**Practical consequence.** When extending the GeoVac framework to a new physical system (nuclear shell model, quantum dot, oscillator basis, screened Coulomb, etc.), the universal sector results may be *quoted* from Paper 22 and Paper 14 without recomputation. Verifying the new system gives the same sparsity is then a single sanity check, not a research program.

## IV. THE POTENTIAL-SPECIFIC SECTOR: TWO RIGIDITY THEOREMS

The potential-specific sector contains results that depend on the explicit form of $D$. The two most consequential examples in the GeoVac framework to date are the Fock projection rigidity theorem (Paper 23) and the Bargmann–Segal rigidity theorem (Paper 24). Both are uniqueness theorems: each identifies a single central potential as the unique system whose spectrum arises from a specific sphere-quotient construction. The two theorems sit on opposite sides of an operator-order divide that we make explicit in Section V.

### A. Fock projection rigidity

Paper 23 [28] states the following:

**Theorem 1** (Fock projection rigidity, Paper 23)**.** *The $S^3$ Fock projection $p_0 = \sqrt{-2E_n}$ maps a one-particle central-field Hamiltonian onto the free Laplacian on $S^3$ if and only if the spectrum $E_{n\ell}$ is independent of $\ell$ within each $n$. The Coulomb potential $-Z/r$ is the unique central potential exhibiting this accidental degeneracy, as a consequence of its hidden* SO(4) *symmetry (the Runge–Lenz vector). Any deformation $V(r) = -Z/r + \delta V(r)$ with $\delta V \not\equiv 0$ lifts the $\ell$-degeneracy, breaking the conformal equivalence between the radial Schrödinger problem and the $S^3$ Laplace–Beltrami operator (the operator whose harmonics the graph's nodes label).*

The proof [28] amounts to observing that Fock's stereographic construction in momentum space requires the hydrogenic relation $E_n \propto -1/n^2$ with no $\ell$-dependence, which fails for any potential without the SO(4) symmetry of the pure Coulomb problem [7].

**What rigidity rules out.** The theorem forbids the following extensions of GeoVac to non-Coulomb central potentials:

- Fock projection $p_0 = \sqrt{-2E}$ does not map the radial Schrödinger equation onto the free Laplacian on $S^3$.
- The integer eigenvalue spectrum $\lambda_n = -(n^2-1)$ of the continuum $S^3$ Laplace–Beltrami operator (carried on the labels as the diagonal $SO(4)$ Casimir; not a limit of the graph Laplacian) does not coincide with the rescaled energy spectrum.
- The $\kappa = -1/16$ kinetic constant has no analog in the new system.

What rigidity does *not* rule out is the angular machinery: the Wigner $3j$ algebra, the Gaunt integrals, the block structure of the two-body interaction tensor. This is the universal-sector content of Paper 22, and it transfers to any central potential.

### B. Bargmann–Segal rigidity

Paper 24 [29] establishes the structural dual of Theorem 1.

**Theorem 2** (Bargmann–Segal rigidity for the 3D HO, Paper 24; **[CONDITIONAL]** on the imported SU(3)-closure converse)**.** *The three-dimensional isotropic harmonic oscillator $V(r) = \frac{1}{2}m\omega^2 r^2$ is the unique central-field potential whose bound-state spectrum arises from the Euler operator on the Hardy space $H^2(S^5)$ restricted to the symmetric irreducible representations $(N, 0)$ of* SU(3)*.*

The proof has three classical inputs [29]: Jauch–Hill [11] (the 3D isotropic HO has SU(3) dynamical symmetry; the converse, that SU(3) closure forces $V \propto r^2$, is imported and not confirmed in Paper 24); Borel–Weil [12] (the symmetric $(N, 0)$ irreps coincide with the homogeneous degree-$N$ subspaces of $H^2(S^5)$); and Bargmann–Segal–Hall [9, 10] (the unique unitary equivalence between $L^2(\mathbb{R}^3)$ and the Bargmann–Fock space, mapping the HO Hamiltonian to $\hbar\omega(\hat{N} + 3/2)$).

**The structural dual.** The two rigidity theorems describe analogous but distinct constructions. We tabulate the parallels in Table I.

The two theorems are not connected by any direct mathematical bridge (there is no central potential that satisfies both simultaneously). What they share is a structural form: each pairs a unique potential with a unique projection construction, and each forbids any modification of the potential without breaking the construction.

### C. The shared structure: rigidity as $D$-dependence

Both rigidity theorems are statements about $D$. They say nothing about $\mathcal{A}$: the algebra of functions on the graph, the spherical-harmonic basis decomposition, the Gaunt selection rules remain valid in either case. What they say is that the specific form of $D$ (the spectrum, the projection used to extract physical energies, the calibration constants) is uniquely determined by the choice of central potential. Two different central potentials produce two different spectral triples that share an algebra but differ in $D$.

This is the formal sense in which rigidity belongs to the potential-specific sector. Any framework result that depends on the explicit form of $D$ — the integer spectrum, the calibration $\kappa = -1/16$, the $\alpha$-combination Observation $K = \pi(B + F - \Delta)$ [18] — is bound to one specific central potential and does not transfer.

TABLE I. The two known central-potential rigidity theorems in the GeoVac framework. Each identifies a unique central potential matching a specific sphere-quotient construction.

| Feature | Coulomb (Paper 23) | 3D HO (Pa |
|---|---|---|
| Sphere | $S^3$ | $S^5$ |
| Geometry | Riemannian | Complex (H |
| Symmetry | SO(4) | SU(3) |
| Operator | 2nd-order Laplacian | 1st-order E |
| Spectrum | $-1/n^2$ (quadratic) | $N + 3/2$ (li |
| Projection | $p_0 = \sqrt{-2E}$ (nonlinear; linear at fixed $p_0$) | linear-affine |
| Where $\pi$ enters | measure, spectral sums; $\kappa$ matched, rational | measure, sp |

## V. WHERE $\pi$ ENTERS

The two rigidity theorems differ in more than the choice of sphere, but not in where $\pi$ enters. Both label sets are $\pi$-free, and above them $\pi$ enters both sides in the same two ways, through a measure and through spectral sums. The operator-order divide of Table I does not separate them.

### A. Operator order and projection type

The Coulomb construction uses the second-order Laplace–Beltrami operator $\Delta_{S^3}$ on the unit three-sphere, with eigenvalues $\lambda_n = -(n^2-1)$, $n \geq 1$ (quadratic in $n$; the same convention as Sec. I, equivalently $-k(k+2)$ with $k = n-1$). The Fock projection $p_0 = \sqrt{-2E_n}$ extracts the physical energy $E_n = -1/(2n^2)$ by matching $p_0$ to the radial scale of momentum-space wavefunctions on $S^3$. This projection is *nonlinear* in the eigenvalue:

$$E_n = -\tfrac{1}{2n^2} \implies E_n^{-1/2} = \sqrt{-2E_n} \cdot n = p_0(n) \cdot n,$$

and the inverse relation $n \mapsto E_n$ involves an inverse square root, and no $\pi$. The matched kinetic constant $\kappa = -1/16$ is also $\pi$-free, coinciding with the geometric $1/16$ as an Observation [19, 29].

The harmonic oscillator construction uses the first-order Euler operator $\hat{N}$ on the Hardy space $H^2(S^5)$, with eigenvalues $N$ (linear in the holomorphic degree). The Bargmann–Segal–Hall projection is unitary and *linear-affine* in the eigenvalue:

$$E_N = \hbar\omega(N + 3/2),$$

and the inverse relation $N \mapsto E_N$ involves no square roots, no rescalings, and no Riemannian volume factors. The only $\pi$ in the Bargmann–Segal construction is the Gaussian normalization $\pi^{-3}$ of the continuous Bargmann measure, which the discrete graph never invokes.

The projection type depends on which number is held fixed. At fixed $p_0$ the Coulomb Sturmians have charges $Z_n = n\,p_0$, linear in the label with one universal constant; in the Kustaanheimo–Stiefel frame they are the constrained states of a four-dimensional oscillator, and the lift costs one fibre volume, $\pi/4$, which cancels in every ratio (Paper 24 [29]).

### B. Spectral sums carry $\pi$ on both sides

Even powers $\pi^{2k}$ enter through spectral zeta values, through $\zeta_R(2k)$, a rational multiple of $\pi^{2k}$, and this holds on both sides. Paper 24 [29] gives the oscillator ladders in closed form: with $g_N = (N+1)(N+2)/2$ the energy ladder $\sum_N g_N(N+\frac{3}{2})^{-4}$ is $\pi^2/4 - \pi^4/48$, beside the $S^3$ Casimir ladder's $\pi^2/12 + 1/16$ at $s = 2$. What the operator order governs on $S^3$ is which zeta values appear, the operator-order axis of the Paper 18 taxonomy [26]: the Dirac spectral-sum work (Paper 28 [31]) shows, via the T9 theorem, that the spectral zeta function of the squared Dirac operator $D^2$ is a two-term polynomial in $\pi^2$ with rational coefficients at every integer $s$ (operator order $2 \to \pi^{\text{even}}$). The parity discriminant theorem of Paper 28 confirms the dual: odd-order operators produce odd-zeta content $\zeta_R(3), \zeta_R(5), \ldots$ via half-integer Hurwitz reduction.

### C. Where $\pi$ does and does not appear

We tabulate which framework results carry $\pi$ in Table II.

**Implication for nuclear extensions.** Paper 23's nuclear shell model qubit Hamiltonians use the harmonic oscillator basis with the Minnesota nucleon–nucleon potential [28]. Paper 23's construction works directly in the HO basis without a stereographic projection: every coefficient is a rational combination of Minnesota Gaussians, Moshinsky–Talmi brackets, and HO matrix elements; no transcendental coefficient appears.

TABLE II. Where $\pi$ appears in GeoVac results. On both potential-specific sides the label-level data are $\pi$-free and $\pi$ enters through measures and spectral sums. Universal-sector results (angular sparsity, $jj$-coupled selection rules) are $\pi$-free, because they involve no projection.

| Result | $\pi$? |
|---|---|
| *Universal sector (algebra only):* | |
| Angular sparsity [27] | no |
| Composed Pauli scaling [23] | no |
| Wigner $3j$ algebra | no |
| *Coulomb ($S^3$):* | |
| Label set $(n,l,m)$, Casimir $n^2-1$ [19] | no |
| $\kappa = -1/16$ [19] | no ($-1/16$ rational; $\pi$ enters at tl |
| Spectral zeta $\zeta_{D^2}(s)$ [31] | yes ($\pi^{2k}$) |
| Volume-slot measure $\mathrm{Vol}(S^2)/4$ [26] | yes |
| $\alpha$-combination Observation $K$ [18] | yes |
| *HO ($S^5$ Hardy):* | |
| Bargmann–Segal graph adjacency [29] | no |
| HO spectrum $\hbar\omega(N+3/2)$ | no |
| Bargmann transition amplitudes | no |
| Bargmann measure $\pi^{-3}$ [29] | yes (never used by the graph) |
| Oscillator spectral zeta at even $s$ [29] | yes ($\pi^{2k}$) |

## VI. THE LAYER SHARPENING (SPRINT 5)

The Sprint 5 investigation of $S^5$ gauge structure (Paper 24 [29]) sharpened the Coulomb/HO asymmetry from two layers (spectrum-computing and calibration $\pi$) to three discriminators as originally reported, the third being Wilson gauge with natural matter coupling. The spectrum-computing layer (2026-09-25, Layer 1 below) and the calibration-$\pi$ layer (2026-10-19, Layer 2) are withdrawn, so one standing discriminator remains. We summarize the potential-specific layers here as the predictive checklist for extending the framework to any new central potential (the full asymmetry is catalogued in Paper 24 [29]: seven layers catalogued, five standing).

### A. The structural layers

**Layer 1: Spectrum-computing role of the node Laplacian $L$**

This layer was catalogued on the premise that, in the Coulomb $S^3$ construction, the node Laplacian $L_0 = D - A$ on the Fock graph is conformally identified with the continuum $S^3$ Laplace–Beltrami [retracted 2026-09-25: graph-laplacian-converges-to-s3] operator and is the kinetic operator (under the $\kappa$ calibration). The premise is false: $L_0$ preserves $l$, splits into $n_{\max}$ disconnected blocks and does not converge to $\Delta_{S^3}$, whose spectrum $\lambda_n = -(n^2-1)$ is carried exactly on the labels by the diagonal $SO(4)$ Casimir (Paper 7). On $S^3$ as in the HO $S^5$ Bargmann–Segal construction — where

the spectrum is read off from the holomorphic degree $N$ of each node, and $L_0$ encodes *dipole transition amplitudes* — the spectrum is diagonal on the labels and $L_0$ is spectrally inert. This is therefore not an asymmetry (Paper 24).

**Layer 2: Calibration $\pi$ (withdrawn 2026-10-19).:** This layer was catalogued on the premise that the Coulomb side requires calibration $\pi$ and the HO side does not. Both label sets are $\pi$-free, and $\pi$ enters both sides through a measure and through spectral sums at even argument; the projection type behind the premise depends on which number is held fixed (Section V; Paper 24). This is therefore not an asymmetry.

**Layer 3: Wilson gauge with natural matter coupling.:** The Sprint 5 result on $S^5$ gauge structure (Paper 24 [29]) showed that the abelian U(1) Wilson–Hodge structure transfers verbatim to the Bargmann–Segal graph (signed incidence, edge Laplacian $L_1 = B^T B$, Hodge decomposition, all hold). Pure-gauge SU(3) Wilson on the Bargmann–Segal graph is also well-defined (Sprint ST-SU3, Paper 24 [29]), but coupling it to the natural $(N, 0)$ matter is not: transitions between $(N, 0)$ and $(N + 1, 0)$ irreps are Clebsch–Gordan intertwiners, not group elements. On the Coulomb side, $S^3 = \mathrm{SU}(2)$ and the spin-1/2 matter shares the representation of the SU(2) Wilson links of Paper 30 — an argument, not yet realised as a coupled construction (Paper 30, open problems). The combinatorial vocabulary of Wilson–Hodge transfers to both spheres, but only on $S^3$ does the matter representation match the link group; on $S^5$ matter coupling needs a Clebsch–Gordan intertwiner. This layer rests on representation theory, not on $L_0$.

### B. The layer table

The features above the divide in Table III are the potential-specific layers — the one standing discriminator established here (row 3; rows 1 and 2 are withdrawn) together with two further layers catalogued in Papers 24 and 51; the features below are the universal-sector features that hold on either sphere.

TABLE III. Coulomb/HO partition by structural layer. Rows 1–3 are the discriminators reported here (Sprint 5 sharpening, Paper 24 [29]); rows 1 and 2 are withdrawn (2026-09-25, 2026-10-19), so row 3 is the one standing discriminator. Rows 4–5 are further potential-specific layers from Papers 24 and 51 [29, 30], part of the asymmetry catalogued in Paper 24 [29] (seven catalogued, five standing). ✓ means the feature is present; "–" means it is absent or not natural.

| Feature | |
|---|---|
| $L_0$ computes spectrum (withdrawn 2026-09-25) | |
| $\pi$ in measures and spectral sums (withdrawn as a discriminator) | |
| Wilson gauge with natural matter coupling | ✓ |
| Posited-wedge-generator Pythagorean (Paper 24 §V) | |
| Two-term spectral action (Paper 51 [30], Cor. 4.2) | ✓ (conf |
| Abelian U(1) Wilson–Hodge | |
| Angular sparsity | |
| Block-diagonal ERI | |

### C. Why the discriminators are independent

The third layer is structurally distinct from the first two. Layer 1 (spectrum-computing) is a question about *what $L_0$ does*; it is determined by the projection rule (Fock vs. Bargmann–Segal). Layer 2 (calibration $\pi$) was a question about *which scalars appear*; its answer is the same on both sides (Section V). Layer 3 (Wilson gauge with natural matter coupling) is a question about *whether the matter shares the representation of the link group*; it is determined by the representation theory of the ambient symmetry group (whether transitions between irreps are SO(4)-equivariant or SU(3)-CG-intertwined), not by $L_0$ and not by operator order. Three structural questions were catalogued; none reduces to either of the others. (Layer 1 is withdrawn as of 2026-09-25: its answer is the same on both spheres, since $L_0$ on $S^3$ does not converge to the $S^3$ Laplacian and is spectrally inert there too; Layer 2 is withdrawn as of 2026-10-19 for the same reason; one standing discriminator remains, Layer 3.)

**Predictive consequence.** For any candidate new central potential, the layer table provides a checklist for what features of the GeoVac framework can be expected to transfer. A new potential matching the Coulomb pattern (e.g., a hidden SO(4)-like symmetry, a stereographic-style projection, integer spectrum) inherits the standing Coulomb-specific discriminator. A potential matching the HO pattern (e.g., a hidden SU($n$)-like symmetry, a complex-analytic projection, linear spectrum) inherits its HO-side counterpart. Any other potential inherits the universal sector but neither set of specific layers, and must be analyzed system by system.

## VII. IMPLICATIONS

The partition framework simplifies the question of how to extend GeoVac to a new system. We list the practical implications.

### A. What transfers for free

Any new central-potential extension of GeoVac inherits the universal sector without re-derivation:

- Angular sparsity bounds at every $l_{\max}$ (Eq. (1)) [27]. At $l_{\max} = 3$, the new system's two-body Hamiltonian is at least 93.94% sparse (global-$M_L$ density $D = 6.06\%$) before any potential-specific feature is invoked.

- Composed-block exactly $Q$-linear Pauli count [23] ($N_{\text{Pauli}} = 27.90 \times Q$ at fixed basis). The structural sparsity pattern behind the composed architecture's advantage over Gaussian baselines (51× to 317× at equal qubit counts for the Coulomb library) carries over to any spherical-fermion system.

- Selection rules for transitions, oscillator strengths, multipole moments, and the Wigner–Eckart content of any spherical tensor operator.

- Block-diagonal structure of two-body interaction tensors in the composed architecture [25].

- $jj$-coupled spinor angular sparsity for relativistic extensions [23] §V.

These features hold by virtue of the algebra $\mathcal{A}$ and the spherical-harmonic basis decomposition; they are independent of the choice of $D$ and therefore independent of the radial potential.

### B. What requires per-system rederivation

Every result that depends on the explicit form of $D$ must be re-derived for the new system:

- The continuum $S^3$ spectrum $\lambda_n = -(n^2 - 1)$, carried on the labels by the diagonal $SO(4)$ Casimir (not a limit of the graph Laplacian, which does not converge to it; no analog unless the system has SO(4) symmetry).

- The kinetic calibration $\kappa$ (no analog of $-1/16$ unless the projection is Fock-style).

- The transcendental content of spectral invariants [26, 31]. The Coulomb spectral zeta function lives in the $\pi$-polynomial ring (Paper 28 T9 theorem); other systems may live in different rings (e.g., the HO Euler ladder at $s = 4$ also carries $\zeta(3)$, Paper 24).

- The fine-structure-constant Observation $K = \pi(B + F - \Delta)$ [18]. This is a Coulomb-specific observation built from $S^3$ spectral data and has no analog on $S^5$ or other sphere-quotients. Specifically, the spectral geometry of $S^5$ is one of Paper 2's eliminated mechanisms (no $\pi^2$ beside a rational), and the $F$-analog candidate $(1/2)[\zeta_R(4) + 3\zeta_R(5) + 2\zeta_R(6)]$ is mixed even/odd zeta with no $\pi^2/6$ factor (Paper 24 [29]).

- Specific selection rules requiring the explicit form of the Dirac operator (e.g., the selection rules of the physical vector-photon vertex, computed from Spin(4) representation theory in Paper 33 [32], are an $S^3$ result; the $S^5$ analog would have to be worked out separately).

### C. The parametric / central sector (Track TS-C, 2026-05-04)

The binary $\mathcal{A}$-versus-$D$ split is, on closer inspection, slightly too coarse. There exists a class of $D$-deformations that are not Coulomb-specific because they live in the *centre* of the operator algebra acting on $\mathcal{H}$. Concretely, deformations of the form

$$D \mapsto D + c \cdot \not{1}_{\mathcal{H}}, \quad c \in \mathbb{R},$$

or additive shifts of the squared spectrum $D^2 \mapsto D^2 + c\cdot\not{1}$, leave the metric data unchanged: self-adjointness (for real $c$), the bounded commutators $[D + c\not{1}, a] = [D, a]$, and the order-one condition are all preserved. They do *not* preserve every Connes axiom. An even grading is broken, since $\{\gamma, D + c\not{1}\} = 2c\,\gamma \neq 0$ for $c \neq 0$ (relevant to Paper 32's chirality-doubled model truncation, which is even; the true Camporesi–Higuchi window is odd); and because $J$ is antilinear, $Jc = \bar{c}J$, so the real-structure relation $JD = \varepsilon DJ$ survives the shift only for real $c$ and only when $\varepsilon = +1$. Such deformations are *formally D-touching but functionally universal*: they support the introduction of a physical scalar parameter without engaging any potential-specific structure.

The two charter members of this central tier are:

- **Rest-mass** (Paper 34 projection 14, Paper 35). The Klein–Gordon and squared-Dirac dispersion relations shift as $\omega^2 \to \omega^2 + m^2$, equivalently $D^2 \to D^2 + m^2 \cdot \not{1}$. The deformation is central by construction, and rest-mass is therefore *universal across central potentials* in the same sense as the angular sparsity theorem.

- **Drake–Swainson normalization of Bethe-log sums** (Paper 34 projection 13). The Bethe-log spectral sum is divided by the fixed rational $2Z^4/n^3$ for every $l$ (with a factor $2l+1$ when the initial magnetic substates are summed; Paper 36 § 5): a scalar rescaling, central by construction, that introduces no variable.

The natural reading of the framework's $\mathcal{A}/D$ split is therefore three-tier rather than two-tier:

1. **Universal sector** ($\mathcal{A}$-only): angular sparsity, composed scaling laws, selection rules independent of $D$.

2. **Parametric / central sector** ($\mathcal{A} + Z(\mathcal{B}(\mathcal{H}))$): rest-mass shifts, Bethe-log normalizations. Universal across central potentials, but parametric in a scalar.

3. **Potential-specific sector** (substantive $D$): rigidity theorems, the projection, transcendental content, the $\alpha$-combination Observation. Per-system rederivation required.

This refinement is descriptive, not predictive: it sorts existing results into three bins and does not change any of the theorems of Sections III or IV. Its value is clarifying: it explains why rest-mass and Drake–Swainson behave as universal-sector projections at the Paper 34 V/D/T level despite formally touching $D$.

### D. What is currently unknown

Several extensions are not yet placed in the partition framework:

- Non-central potentials (multiparticle interactions with explicit angular dependence). The angular sparsity theorem assumes the two-body interaction is rotationally invariant; for non-central interactions the universal sector may shrink.
- Rank-2 tensor interactions (e.g., Breit retarded interactions, spin-orbit, dipole–dipole). These respect the Gaunt selection rules for their angular content but require additional reduced matrix elements; whether the universal sector remains the same is established in Paper 22 §V (spinor extension) for the $jj$-coupled relativistic case but not for general rank-2 tensor interactions.
- Yang–Mills sectors with matter content (Paper 30 [33] is purely gauge; matter-coupled extensions have not been classified). The Coulomb-specific Layer-3 feature is Wilson gauge with *natural matter coupling* (argued on $S^3$, not yet constructed); pure-gauge SU(3) Wilson is well-defined on the HO side (Sprint ST-SU3; Paper 24 [29]), and whether natural $(N, 0)$ matter can be coupled to it without a Clebsch–Gordan intertwiner is an open question.
- Signature extension to Lorentzian $(3, 1)$ via the Bizi–Brouder–Besnard 2018 [38] classification of pseudo-Riemannian spectral triples. This axis is structurally parallel to the central-potential partition and is treated in Section VIII, which classifies Paper 34's twenty-eight projections by their transfer behavior under the Riemannian $(s, t) = (3, 0) \to$ Lorentzian $(s, t) = (3, 1)$ extension ($s$ space, $t$ time dimensions).

## VIII. THE SIGNATURE PARTITION (SPRINT L0, MAY 2026)

The $\mathcal{A}/D$ partition organizes GeoVac results across *central potentials* at fixed Riemannian signature $(s, t) = (3, 0)$ (three space, no time dimensions). In the Bizi–Brouder–Besnard classification [38], which labels a signature by $(m, n) = (t + s,\ t - s)$ mod 8 (their p. 3, arXiv v2 numbering), this is $(m, n) = (3, 5)$ ($n = -3$ mod 8; this is their formula extended formally to odd $m$, since their tables cover even $m, n$ only; Connes' KO-dimension 3 corresponds), and the Lorentzian signature $(s, t) = (3, 1)$ is $(m, n) = (4, 6)$ (their Table 3). Below, bare pairs $(3, 0)$ and $(3, 1)$ denote $(s, t)$. A structurally parallel partition organizes the same results across *signatures*: which features of the framework transfer cleanly from Euclidean $(3, 0)$ to Lorentzian $(3, 1)$, and which require Krein-space machinery, Lorentzian-native heat-trace formalism, or a Lorentzian analog of the van Suijlekom state-space Gromov–Hausdorff distance used in Paper 38 [36] for WH1's commutative-factor convergence result. The signature partition is the structural sibling of the central-potential partition that is the main topic of this paper; both partitions arise from the same Connes spectral-triple data $(\mathcal{A}, \mathcal{H}, D)$.

The signature partition was first articulated in the Sprint L0 audit of May 2026 [37], motivated by the firing of a named literature trigger: the Nieuviarts 2025 [39] twisted-spectral-triple construction provides a morphism (Riemannian twisted spectral triple) $\to$ (pseudo-Riemannian spectral triple) via a Krein-product / twist link, presented as an alternative to Wick rotation without complex numbers. Whether the construction applies cleanly to GeoVac $S^3 = \mathrm{SU}(2)$ is the parallel L2-Nieuviarts-scoping pass and is not assumed here.

### A. The four-bucket classification

Paper 34's twenty-eight projections [35] are classified by the L0 audit into four buckets according to how each projection transfers under the Riemannian $(3, 0) \to$ Lorentzian $(3, 1)$ extension:

**transfers freely.:** Signature-blind: the projection lives on the $V$ (variable) or $D$ (dimension) axis of Paper 34 § III only, with no Riemannian volume form, no Mellin transform of $\mathrm{Tr}\, e^{-tD^2}$, and no Latrémolière propinquity. Holds identically at $(3, 0)$ and $(3, 1)$. Seventeen of the twenty-eight projections sit here: the SU(2) angular machinery (Wigner $3j$, Wigner $D$, multipole / Gaunt termination, bipolar harmonic / Drake combining, Young-tableau symmetry projection), the parameter-only mechanisms (rest-mass, Drake-Swainson, Phillips-Kleinman, Sturmian reparameterization), the consumption of external nuclear-structure scalars from QCD (Foldy / Friar charge density, Zemach magnetization, tensor multipole), the algebra-side molecular and coupled-channel projections (molecule-frame hyperspherical, adiabatic / Born-Oppenheimer, coupled-channel / adiabatic curve, Bargmann-Segal), and the state-side

complement of the dictionary (apparatus identity / von Neumann entropy, which is PSLQ-disjoint from the master Mellin engine per Sprint TD Track 5). These are signature-blind by virtue of $\mathcal{A}$ and the angular decomposition of $\mathcal{H}$; they extend Paper 22's universal sector to the signature axis.

**wick-map free.:** Reduces to the M1 sub-mechanism of the master Mellin engine via the four-witness Wick-rotation theorem [41–44] codified in Sprint Unruh-pendant (CLAUDE.md §2, 2026-05-10): Hawking, Sewell, Bisognano-Wichmann, and Unruh are four faces of one published bridge $\beta = 2\pi/(\text{surface gravity})$ with $2\pi$ the same $\mathrm{Vol}(S^1)$ factor in every case. Any framework observable whose Lorentzian content reduces to compactifying a single Euclidean time circle inherits a Lorentzian reading at no structural cost at the metric-functional level; literal operator-system identification is the named Sprint L1 target (computing the modular Hamiltonian $K$ on $\mathcal{T}_{n_{\max}}$ of a wedge-restricted Camporesi–Higuchi state, then verifying $\sigma_{i\cdot 2\pi} = \text{identity}$ at finite $n_{\max}$; still open, Paper 42 O0 — the finite $K = 2J_z$ of Paper 42 is posited, and its period closure is automatic). Four of the twenty-eight projections sit here: the observation / temporal-window projection of Paper 34 § III.15, the vector-photon promotion of Paper 34 § III.11, the gauge-choice projection of Paper 34 § III.26, and the Wick-rotation projection of Paper 34 § III.27 itself (which is the bridge projection by construction).

**euclidean specific.:** Requires Krein-space machinery (Hilbert space replaced by Krein space carrying fundamental symmetry $\mathcal{J}^2 = I$), Lorentzian-native heat-trace formalism, or a Lorentzian analog of the Latrémolière propinquity. Named blockers below. Five of the twenty-eight sit here: Fock conformal projection (Paper 34 § III.1, requires pseudo-Riemannian sphere quotient), Hopf bundle (Paper 34 § III.2, requires pseudo-Riemannian fibration), stereographic / conformal coordinate change (Paper 34 § III.4, Lorentzian conformal compactification differs in factor structure), Connes-Chamseddine spectral action (Paper 34 § III.6, $\mathrm{Tr}\, f(D^2/\Lambda^2)$ is Euclidean by construction), and Camporesi-Higuchi spinor lift (Paper 34 § III.7, Riemannian spin manifold; Krein spinor at $(3,1)$ not yet constructed).

**mixed.:** Parts transfer freely, parts require extension. Two of the twenty-eight: Wilson plaquette (Paper 34 § III.10, the combinatorial vocabulary — signed incidence, plaquettes, edge Laplacian $L_1 = B^T B$ — transfers freely, but the Marcolli–van Suijlekom = Yang–Mills correspondence is Euclidean), and two-body Dirac / Breit retardation (Paper 34 § III.16, the angular content transfers via Wigner $3j$, but the kinematic content requires Lorentzian re-derivation, though the result remains in $\mathbb{Q}(\alpha)$).

The full per-projection table with one-line mechanism / blocker descriptions is in Paper 34 § V.E [35]; the machine-readable classification with confidence ratings is in `debug/data/lorentzian_l0_audit.json`.

### B. Named blockers

The signature partition's potential-specific side requires extension work on three named blockers, with literature support to varying degrees:

**B1.:** *Lorentzian propinquity.* The van Suijlekom state-space Gromov–Hausdorff distance used in Paper 38 [36] for WH1's commutative-factor convergence result—and the Latrémolière propinquity family more broadly—is purely Riemannian as of May 2026 (Latrémolière, Toyota, Hekkelman–McDonald, Leimbach–van Suijlekom). Constructing a Lorentzian propinquity is original NCG-mathematics work at the long-range scale. The Nieuviarts 2025 trigger-firing [39, 40] may shorten this budget significantly via the twist-morphism mechanism if the construction applies to $S^3 = \mathrm{SU}(2)$ (odd-dimension caveat to verify in Sprint L2).

**B2.:** *Krein-space spectral triple at signature* $(s,t) = (3,1)$*, BBB* $(m,n) = (4,6)$. Bizi–Brouder–Besnard 2018 [38] provides the most concrete prescription: Hilbert space replaced by Krein space carrying fundamental symmetry $\mathcal{J}^2 = I$; sign-table replacement on Connes axioms (signature-dependent $\{J, D, \gamma\}$ anticommutations per $(m,n)$ entry). This is the primary structural extension required for the EUCLIDEAN SPECIFIC bucket.

**B3.:** *Lorentzian-native heat-trace formalism.* Connes–Chamseddine spectral action $\mathrm{Tr}\, f(D^2/\Lambda^2)$ is Euclidean by construction; no published Lorentzian sequel exists. The temporal-circle special case is handled by Wick rotation (clean Sprint TD Track 4 reproduction of Hawking temperature), but general Lorentzian heat-trace evaluation remains an open structural extension.

### C. What transfers freely under signature extension

The combined leverage of the central-potential partition and the signature partition is striking: *the universal sector of the $\mathcal{A}/D$ partition* (Section III) and the TRANSFERS FREELY bucket of the signature partition (Section VIII A) have substantial overlap. Both contain the SU(2) angular sparsity theorem, the composed-block exactly $Q$-linear Pauli count, the Wigner $3j$ / $D$

algebra, and the selection rules. Adding the signature axis to the partition framework reveals that these results transfer not only across central potentials at fixed signature (the composed Pauli coefficient only as expected, Paper 22 Corollary 1) but also across signatures themselves. The framework's structural-skeleton-scope reading (CLAUDE.md §2 sprint-of-2026-05-07 pattern crystallization) survives the $(3,1)$ extension at 75% of the projection dictionary (TRANSFERS FREELY + WICK-MAP FREE, 21 of 28).

### D. The predicted trivial closure at $(3,1)$ (tested: falsified)

The Sprint L0 audit drew a structural prediction from the BBB sign table at Lorentzian signature $(s,t) = (3,1)$, i.e. BBB $(m,n) = (4,6)$: that the M3 sub-mechanism of the master Mellin engine (Dirichlet-$L$ content of the parity-split Camporesi-Higuchi spectral sums [31]) becomes structurally empty on chirality-symmetric spectrum, because the BBB signs there (their Table 1 and eq. (3)) make $J$ anticommute with the chirality, which was expected to remove the asymmetry that drives the half-integer-Hurwitz content of M3. That would have reduced the master Mellin engine's three-mechanism partition (Paper 32 § VIII case-exhaustion theorem) to two on the Lorentzian side. *Tested and falsified (Sprint L2-D; Paper 42 § 11.8, Paper 34):* under the $n_{\mathrm{Fock}}$-parity convention that accesses Paper 28's M3 content, the parity-split difference on the chirality-symmetric $(3,1)$ truncation is bit-identical to the Riemannian value, so M3 does not trivialise; under a chirality-pairing convention the difference vanishes only tautologically and does not access M3. M3 is sectional in $n_{\mathrm{Fock}}$-parity, not in spacetime signature, and the case-exhaustion classification is unchanged.

### E. Sequencing recommendation

The audit's per-bucket classification suggests a natural four-sprint sequence:

**Sprint L0 (sprint-scale, this).:** Documentation only. The seventeen TRANSFERS FREELY projections inherit Section III cleanly; defer to Paper 34 § V.E for the full per-projection table. Status: this section + Paper 34 § V.E + Paper 32 § VIII.E = closure for the leaves.

**Sprint L1 (beyond sprint scale, MEDIUM-HIGH priority).:** Operator-system literal-identification of the Wick-rotation projection (Paper 34 § III.27). Falsifier: compute modular Hamiltonian $K$ on $\mathcal{T}_{n_{\max}}$ for a Rindler-wedge-restricted Camporesi-Higuchi state, verify $\sigma_{i\cdot 2\pi}$ = identity at finite $n_{\max}$, take the GH limit using Paper 38's lemmas [36]. *Does not* require B2; the operator-system extension is internal to the existing Riemannian state-space GH machinery (propinquity is not claimed). One operator-level proof lifts all four Wick-rotation faces simultaneously via the shared KMS-Tomita-Takesaki algebra. *Status (2026-09-28):* open. Paper 42 posits the generator $K = \mathrm{diag}(2m_j)$ and builds the state from it, so its $2\pi$ closure is automatic and does not run this falsifier.

**Sprint L2 (a substantial program).:** BBB Krein lift and Connes axiom audit at $(3,1)$. Gates: blocker B2. Refines the five EUCLIDEAN SPECIFIC rows and the two MIXED rows to definitive classifications. Predicted M3 trivialization on chirality-symmetric spectrum (consistency check; tested and falsified, §VIII D).

**Sprint L3 (long-range).:** Lorentzian propinquity construction (blocker B1). Load-bearing only if Sprint L1 / L2 closure requires Lorentzian Latrémolière propinquity to take the GH limit at finite $n_{\max}$. May be shortened significantly by the Nieuviarts 2025 morphism if L2-Nieuviarts-scoping verifies applicability.

This section is descriptive, not predictive: the four-bucket classification sorts existing projections into transfer categories and does not change any of the theorems in Sections III or IV. Its value, like the rest of this paper, is operational: it tells future Lorentzian extensions which framework results may be quoted from prior papers without modification, and which require system-specific computation. Sprint L1 and Sprint L2 will deepen the classification with actual computations; at the present stage of documentation only, this section records the audit verdict and identifies the sequencing.

### F. Bucket classification of the 28 projections by Sprints L1 + L2 (2026-05-16/17; seven exercised at finite cutoff)

The four-bucket classification above was an a priori prediction at the time of Sprint L0 (May 2026). Subsequent sprints have classified all 28 projections into these buckets; seventeen rows hold by construction and four by prior-art inheritance, so only seven are exercised at finite cutoff:

- Seventeen TRANSFERS FREELY projections inherit Section III's universal $\mathcal{A}$-side sector verbatim; they involve no Riemannian volume form, no Mellin transform of $\mathrm{Tr}\, e^{-tD^2}$, and no Latrémolière propinquity, so they are signature-blind by construction.
- Four WICK-MAP FREE projections reduce to the M1 sub-mechanism via the *published* four-witness Wick-rotation theorem (Hartle–Hawking + Sewell + Bisognano–Wichmann + Unruh) at the metric-functional level; that reduction is prior-art inheritance, not a framework-internal verification. Sprint

L1 (Paper 42 §§ 5–8) does *not* realize the theorem at the operator-system level at finite cutoff: its $2\pi$ period closure is automatic for the posited integer-spectrum generator $K = \mathrm{diag}(2m_j)$, the Tomita step is circular, and the four witnesses agree by construction (an earlier reading of it as an identification at the operator-system level is withdrawn, 2026-09-28; deriving $K$ from a Camporesi–Higuchi wedge state is open). The Sprint L2-E Krein period closure (Paper 32 § VIII.E.E) is likewise automatic at finite cutoff — the posited generator is compact ($e^{2\pi i K} = I$ on its integer spectrum; [SYMBOLIC]), so its flow is a $2\pi$ circle. That the Lorentzian signature is therefore metrically invisible at finite cutoff (the Euclidean-only reading) is [ARGUED] and generator-independent; reading the circle as the Wick-rotated KMS $\beta = 2\pi$ circle is conditional on Paper 42's open O0. The strong "literal identification" reading of Sprint L2-E (2026-05-17) was withdrawn 2026-06 (see Paper 34 § III.29 and Paper 45).

- Five EUCLIDEAN SPECIFIC projections were carried to the Krein operator-system level at finite cutoff via Sprint L2-B/C/D/E (Paper 32 § VIII.E.B/C/D/E): the Krein space, Lorentzian Dirac $D_L = i^t \not{D}_{g_r}$ per van den Dungen 2016 Prop 4.1, and Connes axiom audit at BBB $(m, n) = (4, 6)$ are constructed, and the Krein period-closure identities close bit-exactly at every tested $(n_{\max}, N_t) \in \{1, 2, 3\} \times \{1, 11, 21\}$. These identities are *signature-blind*, however (the same compactness mechanism as above; the signature-blindness reading is [ARGUED]), so they do *not* constitute a Lorentzian literal identification; the genuine Lorentzian propinquity extension (blocker B1) remains open at Sprint L3 scope.
- Two MIXED projections (Wilson plaquette #10, Breit retardation #16) preserve their per-case structural distinctions: combinatorial Wilson vocabulary transfers freely but the Marcolli–van Suijlekom Yang–Mills correspondence is Euclidean; Breit angular content transfers via Wigner $3j$ but the kinematic content requires Lorentzian re-derivation while remaining in $\mathbb{Q}(\alpha)$.

The complete classification of the 28 projections is the direct operational payoff of the Sig/Op framing introduced here: it certifies the *bucket classification* (which projections transfer freely, which require system-specific re-derivation), not a Lorentzian closure. The signature partition, like the central-potential partition that is the main topic of this paper, is now a complete classification of GeoVac's projection inventory, seven of whose 28 rows are exercised at finite cutoff. At the operator-system / finite-cutoff level neither closure is established: the *Riemannian* period closure is automatic for a posited generator (the derivation of the wedge modular Hamiltonian from a Camporesi–Higuchi state is open, Paper 42), and the *Lorentzian* closure is additionally blocked — the flow of the posited finite generator $K = 2J_z$ (its Bisognano–Wichmann identification is open, Paper 42 O0) is compact, a $2\pi$ circle [SYMBOLIC]; that Lorentzian signature is therefore metrically invisible at any finite cutoff (the Euclidean-only reading) is [ARGUED] and generator-independent, while the reading of that circle as the Wick-rotated KMS $\beta = 2\pi$ circle is conditional on O0. The genuine Lorentzian extension (Sprint L3, blocker B1) remains open as major NCG-mathematics work (see Paper 34 § III.29, Paper 45). The headline structural finding of the verification chain: Paper 42 § 7.2 "$H_{\mathrm{local}} \neq D_W$" (the contrast between the posited generator and the wedge Dirac) is *signature-independent* at the Riemannian limit, an elementary property of the posited construction rather than evidence about which generator a Camporesi–Higuchi wedge state selects.

## IX. OPEN QUESTIONS

The partition framework as stated is descriptive: it sorts existing results into universal and potential-specific buckets and predicts which results carry over to new systems. It is not yet predictive in the deeper sense of establishing the partition as a theorem. Three open questions follow.

### A. Is the partition exhaustive?

The framework as stated identifies two known potential-specific projections (Fock for Coulomb, Bargmann–Segal for HO) and asserts that they sit on opposite sides of an operator-order divide. Are there other rigidity theorems we have missed? Specifically:

- The Morse oscillator $V(r) = D_e(1 - e^{-a(r-r_e)})^2$ has an exact analytic spectrum with SU(2) dynamical symmetry but is one-dimensional; whether a 3D radial Morse-like potential admits a sphere-quotient discretization with a rigidity theorem analogous to Theorems 1 and 2 is unknown.
- The Pöschl–Teller potential and other shape-invariant potentials [14, 15] have algebraic ladder structure analogous to the HO; whether any of them admit a Bargmann-style discretization on a non-$S^5$ Hardy space is unknown.
- The Coulomb-Sturmian basis [20] is closer to the $S^3$ Coulomb construction but uses a different boundary condition; it has its own selection rules and transcendental structure that have not been placed in the partition framework.

A complete classification of central potentials by sphere-quotient projection type would put the partition framework on firm mathematical footing.

### B. What is the universal/specific partition for non-central interactions?

The universal sector as stated assumes rotationally invariant two-body interactions. For non-central interactions (gauge couplings, spin–orbit, multipole moments coupled to external fields), the angular structure of the interaction itself enters the selection rules. In some cases (e.g., the vector-photon vertex on $S^3$, Paper 33 [32]) the non-central content is captured by a vertex coupling fixed by the representation theory of Spin(4) = $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$. In other cases (Breit retarded interactions in Paper 14 §V [23]) it requires an explicit reduced matrix element that goes beyond the Gaunt algebra. Whether there is a general universal/specific partition for non-central interactions, analogous to the rotationally invariant case, is not yet established.

### C. Does the partition predict $\alpha$?

The fine-structure constant Observation $K = \pi(B + F - \Delta) \approx \alpha^{-1}$ [18] sits in the potential-specific Coulomb sector by the partition framework. The headline observation matches CODATA to $4.8 \times 10^{-7}$ for $K$ itself and to $8.8 \times 10^{-8}$ through the root of the cubic $1/\alpha = K - \alpha^2$ [18], but the combination rule itself remains a numerical observation without first-principles derivation [34]. The spectral geometry of $S^5$ is among the mechanisms Paper 2 tested and eliminated [18], consistent with the partition placing $K$ in the Coulomb-specific sector. Whether the partition framework can be sharpened to *predict* the combination rule, or whether $K$ is genuinely a structural coincidence among three independent Coulomb-side ingredients [18], is the most consequential open question in the framework.

## X. TWO-BODY VERIFICATION OF THE PARTITION

The $\mathcal{A}/D$ partition can be tested directly by constructing the tensor-product spectral triple $\mathcal{T}_1 \otimes \mathcal{T}_2$ for two particles on the Fock-projected $S^3$ and asking what interaction, if any, emerges from the framework's own structure.

### A. Gauged spectral action

The free tensor-product Dirac $D_{\mathrm{total}} = D_1 \otimes I + \gamma_1 \otimes D_2$ satisfies $\{D, \gamma\} = 0$ exactly, so $D^2_{\mathrm{total}}$ factorizes and the free spectral action contains no interaction. Gauging via the inner fluctuation $A = \sum_i a_i^\dagger [D, a_i]$ from the 3-$Y$ multiplier algebra produces a genuine two-body interaction: the connected (non-factorizable) fraction of $\{D, A\}$ is 77% at $n_{\max} = 2$ and 32% at $n_{\max} = 3$.

This interaction is supported only on the direct (density–density) elements $\langle ab| \cdot |ab\rangle$ (25 and 196 nonzeros, against 107 and 3800 for $1/r_{12}$), so it is $m$-conserving, $\Delta l = 0$ and pure $k = 0$ by construction: a strict subset of the Coulomb selection rules, with no exchange or $k > 0$ channel **[MEASURED]**. Its radial matrix elements do not match Coulomb either: on the shared support the Pearson correlation is 0.41–0.58 and decreases with $n_{\max}$.

### B. Resolvent construction

The spectral action uses heat-kernel weights $\mathrm{Tr}\, f(D^2/\Lambda^2)$. An alternative is the resolvent $(D^2)^{-1}$, which in the continuum gives the Green's function. The two-body matrix elements from the resolvent are

$$V^{\mathrm{res}}_{abcd} = \sum_{NLM} M_{NLM}[a, c] \cdot \overline{M_{NLM}[d, b]} \cdot w(N), \qquad (2)$$

where $M_{NLM}$ are the 3-$Y$ multiplier matrices and $w(N)$ is the resolvent weight. Four candidates were tested: the Laplacian resolvent $w = 1/(N^2 - 1)$, the Dirac resolvent $w = 1/(N + \frac{1}{2})^2$, uniform $w = 1$, and $w = 1/N$.

The Dirac resolvent gives the highest Pearson correlation (0.846 at $n_{\max} = 2$, 0.796 at $n_{\max} = 3$, after the 2026-08-29 ERI-evaluator fix; the earlier 0.81/0.75 are retired), outperforming the spectral action. But the correlation *decreases* with $n_{\max}$ across all four weightings. The Laplacian resolvent is essentially uncorrelated ($|r| < 0.05$ at $n_{\max} = 3$, and well below the Dirac resolvent's correlation at every cutoff) because the $N = 1$ zero mode of $\Delta_{S^3}$ excludes the dominant $(1s, 1s)$ contribution. The Dirac operator naturally regularizes this zero mode.

### C. Structural diagnosis

The 3-$Y$ multiplier $M_{NLM}$ factorizes as $I^{S^3}_{\mathrm{rad}} \times G_{\mathrm{Gaunt}}$: the Gegenbauer triple integral on $S^3$ times the $S^2$ Gaunt coefficient. The Coulomb two-electron integral factorizes as $G_{\mathrm{Gaunt}} \times R^k_{\mathrm{Slater}}$: the same Gaunt coefficient times the flat-space Slater integral with the $r^k_</r^{k+1}_>$ kernel.

The **angular** Gaunt part is shared — hence both routes are consistent with the Coulomb selection rules (the spectral action populating only their monopole, direct subset). The **radial** parts are structurally different: the Gegenbauer triple integral on $S^3$ versus the Slater integral in flat Euclidean coordinates. The conversion between them is the Fock projection (Paper 7 [19]), specifically the conformal factor $r_{\mathrm{chordal}} = 2R \sin(\chi_{\mathrm{geodesic}}/2R)$, which is a nonlinear function of the geodesic angle, not a single calibration constant.

This places the multi-focal composition wall — documented across six independent observables in the framework [35] — precisely at the $\mathcal{A}/D$ boundary:

$$V_{12} = \underbrace{\text{angular selection rules}}_{\mathcal{A}\ \text{(universal)}} \times \underbrace{\text{radial coupling strengths}}_{D/\text{metric (potential-specific)}}. \qquad (3)$$

The angular algebra determines which channels may couple (Paper 22); the gauged fluctuation populates only their monopole, direct subset. The Dirac operator — through the Fock projection that maps its eigenvalues to physical coordinates — determines how strongly.

## XI. CONCLUSION

The GeoVac framework is a discrete spectral triple in the graph-spectral-triple lineage of Marcolli and van Suijlekom [3], and its results decompose cleanly along the standard $\mathcal{A}$ versus $D$ split. The universal sector — containing angular sparsity bounds, composed-block Pauli scaling, selection rules, and the $jj$-coupled spinor extension — depends only on the algebra $\mathcal{A}$ and the angular structure of its basis. These results transfer to any spherical-fermion system without re-derivation (the composed Pauli coefficient is expected, not proven, to carry over; Paper 22, Corollary 1). The potential-specific sector — containing the spectrum, the calibration $\kappa$, the transcendental content, the rigidity theorems, and the $\alpha$-combination Observation — depends on the explicit form of $D$. These results must be re-derived for each new system.

The two known potential-specific rigidity theorems (Fock for Coulomb, Bargmann–Segal for HO) sit on opposite sides of an operator-order divide: Coulomb is a second-order Riemannian construction, HO a first-order complex one. The divide does not decide where $\pi$ enters; both sides carry it in a measure and in spectral sums (Paper 24). The Sprint 5 sharpening leaves one standing discriminator, Wilson gauge with natural matter coupling, which is decided by representation theory (on $S^3$ matter and links share a representation, argued but not yet constructed; on $S^5$ matter coupling needs a Clebsch–Gordan intertwiner). Calibration $\pi$ is withdrawn (2026-10-19), and the spectrum-computing role of $L_0$ is [retracted 2026-09-26: coulomb-ho-layer1-and-three-discriminators] withdrawn (2026-09-25: the $S^3$ graph Laplacian does not converge to the $S^3$ Laplacian and is spectrally inert, as on $S^5$). Paper 24 [29] catalogues the full asymmetry as seven structural layers, five standing.

The framing offered by this paper has no new theoretical content; it is a structural reorganization of existing results in spectral-triple language. Its value is operational: it tells future extensions of the framework which results may be quoted from prior papers and which require system-specific computation, and it places the framework in a lineage where the relevant tools (graph spectral triples, gauge networks, spectral truncations) already exist in the mathematical literature.

**Acknowledgement of source material.** Every theorem cited in this paper is established in Papers 22 [27], 23 [28], 24 [29], or 30 [33]. The contribution here is the explicit $\mathcal{A}/D$ partition framing and the predictive checklist following from the layer table (Table III). Numerical verifications of the universal sector are in Paper 22 [27] (five potentials at $l_{\max} = 2$); numerical verifications of the $\pi$-free HO-specific sector are in Paper 24 [29] (Bargmann–Segal lattice at $N_{\max} = 5$, 56 nodes, 165 edges, zero irrationals); numerical verification of the $S^5$ gauge structure non-naturalness is in Paper 24 [29].

[1] A. Connes, *Noncommutative Geometry* (Academic Press, San Diego, 1994).
[2] A. H. Chamseddine and A. Connes, "The Spectral Action Principle," Commun. Math. Phys. **186**, 731 (1997).
[3] M. Marcolli and W. D. van Suijlekom, "Gauge networks in noncommutative geometry," J. Geom. Phys. **75**, 71 (2014); arXiv:1301.3480.
[4] C. I. Pérez-Sánchez, "Bratteli networks and the Spectral Action on quivers," arXiv:2401.03705 (2024).
[5] C. I. Pérez-Sánchez, "Comment on 'Gauge networks in noncommutative geometry'," arXiv:2508.17338 (2025).
[6] A. Lichnerowicz, "Spineurs harmoniques," C. R. Acad. Sci. Paris **257**, 7 (1963).
[7] M. Bander and C. Itzykson, "Group Theory and the Hydrogen Atom," Rev. Mod. Phys. **38**, 330 and 346 (1966).
[8] V. Fock, "Zur Theorie des Wasserstoffatoms," Z. Phys. **98**, 145 (1935).
[9] V. Bargmann, "On a Hilbert space of analytic functions and an associated integral transform, Part I," Commun. Pure Appl. Math. **14**, 187 (1961).
[10] B. C. Hall, "The Segal–Bargmann coherent state transform for compact Lie groups," J. Funct. Anal. **122**, 103 (1994).
[11] J. M. Jauch and E. L. Hill, "On the Problem of Degeneracy in Quantum Mechanics," Phys. Rev. **57**, 641 (1940).
[12] A. Borel and A. Weil, "Représentations linéaires et espaces homogènes kählerіens des groupes de Lie compacts," as exposited by J.-P. Serre, Séminaire Bourbaki, exposé 100 (1954).
[13] K. G. Dyall and K. Fægri, *Introduction to Relativistic Quantum Chemistry* (Oxford University Press, Oxford, 2007).
[14] L. Infeld and T. E. Hull, "The Factorization Method," Rev. Mod. Phys. **23**, 21 (1951).
[15] F. Cooper, A. Khare, and U. Sukhatme, "Supersymmetry and Quantum Mechanics," Phys. Rep. **251**, 267 (1995).
[16] J. Loutey, "Counting Angular Momentum Labels by Isotropic Packing," GeoVac Paper 0 (2026).
[17] J. Loutey, "The Geometric Atom: the Hydrogen State Space as a Label Lattice," GeoVac Paper 1 (2026).
[18] J. Loutey, "The Fine Structure Constant as a Spectral Coincidence on the Three-Sphere," GeoVac Paper 2 (2026). [Observation.]
[19] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation on a Scale-Invariant Label Lattice," GeoVac Paper 7 (2026).
[20] J. Loutey, "Bond Sphere: Diatomic Molecules as $S^3$

and their Natural Sturmian Basis," GeoVac Papers 8–9 (2026).
[21] J. Loutey, "The Hyperspherical Lattice: Two-Electron Atoms as Coupled Channel Graphs," GeoVac Paper 13 (2026).
[22] J. Loutey, "The Molecular Fock Projection: Diatomic Molecules as Prolate Spheroidal Lattices," GeoVac Paper 11 (2026).
[23] J. Loutey, "Structurally Sparse Qubit Hamiltonians from Natural-Geometry Hydrogenic Bases," GeoVac Paper 14 (2026).
[24] J. Loutey, "The Level 4 Natural Geometry: Two-Center Two-Electron Molecules in Molecule-Frame Hyperspherical Coordinates," GeoVac Paper 15 (2026).
[25] J. Loutey, "Composed Natural Geometries for Core-Valence Diatomics: LiH and $BeH^+$ from Natural-Geometry Solvers," GeoVac Paper 17 (2026).
[26] J. Loutey, "Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds," GeoVac Paper 18 (2026).
[27] J. Loutey, "Potential-Independent Angular Sparsity for Qubit Hamiltonians in Spherical Fermion Systems," GeoVac Paper 22 (2026).
[28] J. Loutey, "Nuclear Shell Model Hamiltonians on the Hyperspherical Lattice," GeoVac Paper 23 (2026).
[29] J. Loutey, "The Bargmann–Segal Lattice: $\pi$-Free Discretization of the Harmonic Oscillator on $S^5$," GeoVac Paper 24 (2026).
[30] J. Loutey, "A Convergent Discrete Replica Derivative on the Spinor Cone, with the Two-Term Camporesi–Higuchi Spectral Action on $S^3 \times S^1$," GeoVac Paper 51 (2026).
[31] J. Loutey, "Periods of Dirac Spectral Sums on $S^3$," GeoVac Paper 28 (2026).
[32] J. Loutey, "Selection Rules of the Vector-Photon Vertex on $S^3$: the Physical Vertex, the Vertex Model and the Graph," GeoVac Paper 33 (2026).
[33] J. Loutey, "Wilson Lattice Gauge Theory on the GeoVac Label Graphs: Exact Factorization on the Scalar Graph and Compact $U(1)$ on the Dirac Rule-B Graph," GeoVac Paper 30 (2026).
[34] J. Loutey, *Curve-fit / pattern-match audit memo*, GeoVac internal document, `docs/curve_fit_audit_memo.md` (2026-05-02).
[35] J. Loutey, "GeoVac as a Two-Layer Framework: Projection Taxonomy and the Empirical Matches Catalog," GeoVac Paper 34 (2026).
[36] J. Loutey, "State-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples on $S^3$," GeoVac Paper 38 (2026), Zenodo.
[37] J. Loutey, *Sprint L0 — 28-Projection Lorentzian Transfer Audit*, GeoVac internal document, `debug/lorentzian_l0_audit_memo.md` (2026-05-16).
[38] N. Bizi, C. Brouder, and F. Besnard, "Space and time dimensions of algebras with applications to Lorentzian noncommutative geometry and quantum electrodynamics," *J. Math. Phys.* **59**, 062303 (2018), arXiv:1611.07062.
[39] G. Nieuviarts, "Emergence of Lorentz symmetry from an almost-commutative twisted spectral triple," arXiv:2502.18105 (2025).
[40] G. Nieuviarts, "Emergence of Time from a Twisted Spectral Triple in Almost-Commutative Geometry," arXiv:2512.15450 (2026).
[41] J. B. Hartle and S. W. Hawking, "Path-integral derivation of black-hole radiance," Phys. Rev. D **13**, 2188 (1976).
[42] G. L. Sewell, "Quantum fields on manifolds: PCT and gravitationally induced thermal states," Ann. Phys. **141**, 201 (1982).
[43] J. J. Bisognano and E. H. Wichmann, "On the duality condition for quantum fields," J. Math. Phys. **17**, 303 (1976).
[44] W. G. Unruh, "Notes on black-hole evaporation," Phys. Rev. D **14**, 870 (1976).